

12(a)(i)	<u>nucleus</u> emits particles/EM radiation/ionising radiation	B1
	emission/release from unstable <u>nucleus</u> or emission from <u>nucleus</u> is random and/or spontaneous	B1
12(a)(ii)	probability of decay (of a nucleus) or fraction of (number of undecayed) nuclei that will decay	M1
	per unit time	A1
12(b)	energy is shared with another particle	B1
	mention of antineutrino	B1
12(c)(i)	number = $[(1.2 \times 10^{-9}) / 131] \times 6.02 \times 10^{23}$ or number = $(1.2 \times 10^{-3} \times 10^{-9}) / (131 \times 1.66 \times 10^{-27})$ (= 5.51×10^{12})	C1
	$A = \lambda N$	C1
	= $[0.086 / (24 \times 3600)] \times 5.51 \times 10^{12}$ = 5.5×10^6 Bq	A1
12(c)(ii)	$1/50 = \exp(-0.086t)$ or $1/50 = 0.5^n$	C1
	$t = 45$ days	A1